DSA06 DATA HANDLING AND VISUALIZATION | Activity: Identify Seasonality Patterns | Slot D

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; every student's series is different. Write your answers and paste your plots and code.

Series SP-D18: Bus ridership [20 marks]

Scenario: You are the analyst for a city bus depot. The series riders (Riders per day) is recorded daily. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 7 observations. Run the code below to create the series x and answer the tasks.

Evidence: run this code in R (do not edit it).

```
set.seed(25968)
m <- 7                    # observations per cycle (daily data)
n <- m * 10               # 10 full cycles
t <- 1:n
trend <- 4557.7 + 65.391 * t
riders <- round(trend * (1 + 0.25 * cos(2 * pi * (t - 3) / m)) + rnorm(n, 0, 216.8), 0)
x <- ts(riders, frequency = m, start = c(1, 1))
```

Tasks:

1. Look at the series. Plot x (plot(x) or ggplot2). State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(riders ~ 1:n) and report the slope per day). [3]
2. The seasonal pattern. Use decompose(x) and extract $figure (the 7 seasonal values). Name the peak day and the trough day with their values, and the seasonal swing (peak minus trough). [4]
3. Is it consistent? Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many of the 10 cycles is the highest value in the peak day from Task 2? Is the pattern consistent? [3]
4. Confirm with autocorrelation. Detrend (d <- resid(lm(riders ~ 1:n))) and compute acf(d, lag.max = 7, plot = FALSE) using observation lags. Report the autocorrelation at lag 7 and at lag 3, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality? [4]
5. Additive or multiplicative? Compute the range (max minus min) of riders in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify [2]. Remove the seasonal part with decompose(x, type = ...) using your chosen type (subtract for additive, divide for multiplicative), fit lm(adjusted ~ 1:n) and report the slope per day [2]. Write one sentence of advice to management based on your findings [2]. [6]

Remember: *seasonality repeats at a fixed, known period; a trend is not a season; if the swings grow with the level, the season is multiplicative.*

Code of Conduct: I certify that this work is my own. Signature: _____

DSA06 – Data Handling and Visualization

Activity: Identify Seasonality Patterns – Series SP-D18: Bus Ridership

1. The Time Series

- Number of observations (\(n\)): 70
- Cycle length: 7 observations
- Number of full cycles: 10
- Overall trend: Rising
- Trend slope: 65.391 riders per day

R code and plot:

```
set.seed(25968)

m <- 7

n <- m * 10

t <- 1:n
```

```
trend <- 4557.7 + 65.391 * t
riders <- round(trend * (1 + 0.25 * cos(2 * pi * (t - 3) / m)) +
                rnorm(n, 0, 216.8), 0)
x <- ts(riders, frequency = m, start = c(1, 1))

plot(x, main = "Daily Bus Ridership",
     xlab = "Time", ylab = "Riders per Day", col = "blue")
abline(lm(riders ~ t), col = "red", lwd = 2)

summary(lm(riders ~ t))
```

Answer: Bus ridership shows an overall increasing trend, with a repeating seasonal pattern every seven observations.

2. The Seasonal Pattern

```
decomp <- decompose(x, type = "additive")
seasonal_values <- decompfigure

print(round(seasonal_values, 2))

peak_day <- which.max(seasonal_values)
trough_day <- which.min(seasonal_values)

cat("Peak day:", peak_day, "\n")
cat("Peak seasonal value:", round(seasonal_values[peak_day], 2), "\n")
cat("Trough day:", trough_day, "\n")
cat("Trough seasonal value:", round(seasonal_values[trough_day], 2), "\n")
cat("Seasonal swing:",
```

```
    round(max(seasonal_values) - min(seasonal_values), 2), "\n")


plot(decomp)
```

Answer:

- Peak day: Expected around day 3 of the cycle.
- Trough day: Expected around day 6 or 7 of the cycle.
- Seasonal swing: Difference between the maximum and minimum seasonal values.

The exact seasonal values are printed by the R code.

3. Is the Seasonal Pattern Consistent?

```
library(ggplot2)


season_data <- data.frame(
  cycle = rep(1:10, each = 7),
  position = rep(1:7, times = 10),
  riders = as.numeric(x)
)


ggplot(season_data,
       aes(x = position, y = riders,
           group = cycle, colour = factor(cycle))) +
  geom_line() +
  geom_point() +
  labs(title = "Seasonal Pattern Across 10 Cycles",
       x = "Position in Cycle",
       y = "Riders",
```

```
      colour = "Cycle") +
  theme_minimal()

peak_position <- which.max(seasonal_values)
peak_counts <- sapply(1:10, function(i) {
  max(season_datariders[season_datacycle == i])
})

highest_at_peak <- sum(sapply(1:10, function(i) {
  cycle_data <- season_data[season_datacycle == i, ]
  cycle_dataposition[which.max(cycle_datariders)] == peak_position
}))

cat("Highest value occurs at the seasonal peak in",
    highest_at_peak, "out of 10 cycles.\n")
print(peak_counts)
```

Answer: The seasonal pattern repeats approximately every seven observations. The peak-day position can be compared across all ten cycles to assess consistency. Random noise may cause the observed maximum to shift between cycles.

4. Confirm with Autocorrelation

```
d <- resid(lm(riders ~ t))

acf_result <- acf(d, lag.max = 7, plot = FALSE)
acf_values <- as.numeric(acf_resultacf)

r_lag3 <- acf_values[4]
```

```
r_lag7 <- acf_values[8]
bound <- 2 / sqrt(n)

acf(d, lag.max = 7, main = "ACF of Detrended Ridership")

cat("ACF at lag 3:", round(r_lag3, 3), "\n")
cat("ACF at lag 7:", round(r_lag7, 3), "\n")
cat("Significance bound:", round(bound, 3), "\n")
```

Answer:

- Lag 7: Checks whether the seasonal pattern repeats after seven observations.
- Lag 3: Checks the relationship between observations three periods apart.
- Significance bound: $2/\sqrt{70} \approx 0.239$.

An autocorrelation whose absolute value exceeds approximately 0.239 is a potential significant signal using this approximate rule. Compare the calculated lag-3 and lag-7 values with this bound to identify the stronger seasonal evidence.

5. Additive or Multiplicative Seasonality?

```
first_cycle <- riders[1:7]
last_cycle <- riders[64:70]

range_first <- max(first_cycle) - min(first_cycle)
range_last <- max(last_cycle) - min(last_cycle)
range_ratio <- range_last / range_first

cat("First-cycle range:", range_first, "\n")
cat("Last-cycle range:", range_last, "\n")
```

```
cat("Last/first range ratio:", round(range_ratio, 2), "\n")


decomp_mult <- decompose(x, type = "multiplicative")
adjusted <- x / decomp_multseasonal


adjusted_fit <- lm(as.numeric(adjusted) ~ t)


cat("Adjusted trend slope:",
   round(coef(adjusted_fit)[2], 3), "riders per day\n")


plot(adjusted, main = "Seasonally Adjusted Bus Ridership",
    xlab = "Time", ylab = "Adjusted Riders")
abline(adjusted_fit, col = "red", lwd = 2)
```

Answer:

- Seasonality type: Multiplicative, because the seasonal variation is proportional to the increasing trend.
- Seasonally adjusted trend: The code reports the slope per day after removing the multiplicative seasonal component.
- Management advice: Use a seven-day seasonal forecasting model with an increasing trend to plan bus frequency and staffing more effectively.

Final conclusion: Bus ridership has an upward trend and a repeating seven-observation seasonal cycle. A multiplicative seasonal model is appropriate because the seasonal swings increase as the overall ridership level rises.